

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Automation and Robotics) SEMESTER - 5 Summer 2025 (Supply.)

Course :Bachelor of Technology (Automation and Robotics) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTMOE505D - PRODUCT DESIGN ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

1. What is the main goal of reverse engineering?
 - a) To reduce cost of product
 - b) To understand and replicate existing products
 - c) To increase manufacturing time
 - d) To test software
 2. Disassembling a product helps to:
 - a) Change the product's color
 - b) Understand component relationships and materials
 - c) Sell the product
 - d) Paint the product
 3. Ideation in product design refers to:
 - a) Manufacturing of the product
 - b) Storing data
 - c) Generating and developing ideas
 - d) Buying tools
 4. 3D visualization helps in:
 - a) Cost estimation
 - b) Virtual product understanding
 - c) Creating websites
 - d) Data analysis
 5. Prototype testing is useful to:
 - a) Improve product packaging
 - b) Evaluate the working of a product before final production
 - c) Reduce employee count
 - d) Market the product
- Multidisciplinary teams include:
- a) People from the same department
 - b) Teams with same skillset
 - c) Experts from different fields
 - d) Only managers

7. CAD stands for:
a) Computer Aided Drafting
b) Central Analysis Design
c) Computer Area Division
d) Component Assembly Data
8. Which document provides safe handling instructions for materials?
a) User manual
b) Invoice
c) Safety sheet
d) MSDS (Material Safety Data Sheet)
9. CAM is mainly used for:
a) Material storage
b) Manual calculations
c) Automated manufacturing processes
d) Market surveys
10. Product life cycle includes:
a) Only the production stage
b) Use stage only
c) All stages from design to disposal
d) None of these
11. Recycling of a product means:
a) Throwing it away
b) Reusing parts or materials
c) Hiding the product
d) Selling the waste
12. One of the major goals of environmental concern in design is to:
a) Increase pollution
b) Reduce cost only
c) Increase complexity
d) Minimize environmental impact
- Q2. Solve the following.
- A) Describe the process of idea generation and funneling in the context of product design. 6
- B) Explain the concept of reverse engineering and its importance in product design. 6
- Q3. Solve the following.
- A) Discuss the steps involved in shortlisting ideas from a brainstorming session for product development. 6
- B) What role does product architecture play in the conceptualization of a new product? 6
- Q4. Solve Any Two of the following.
- A) Discuss the benefits and challenges of working in multidisciplinary teams during product development. 6
- B) What is the role of process safety in the testing and evaluation stage of product design? 6
- C) What are the key considerations during prototype testing and evaluation? 6
- Q5. Solve Any Two of the following.

- A) What is the significance of CAD/CAM tools in the preparation of engineering drawings? 6
- B) Explain the importance of the materials and safety data sheet in the manufacturing process. 6
- C) Discuss the importance of manufacturing process identification in design planning. 6
- Q6. Solve Any Two of the following.
- A) What are the major stages of product life-cycle management and why are they important? 6
- B) How can products be designed for easier recycling and reuse? 6
- C) Discuss the environmental implications of improper product disposal. 6

*** End ***